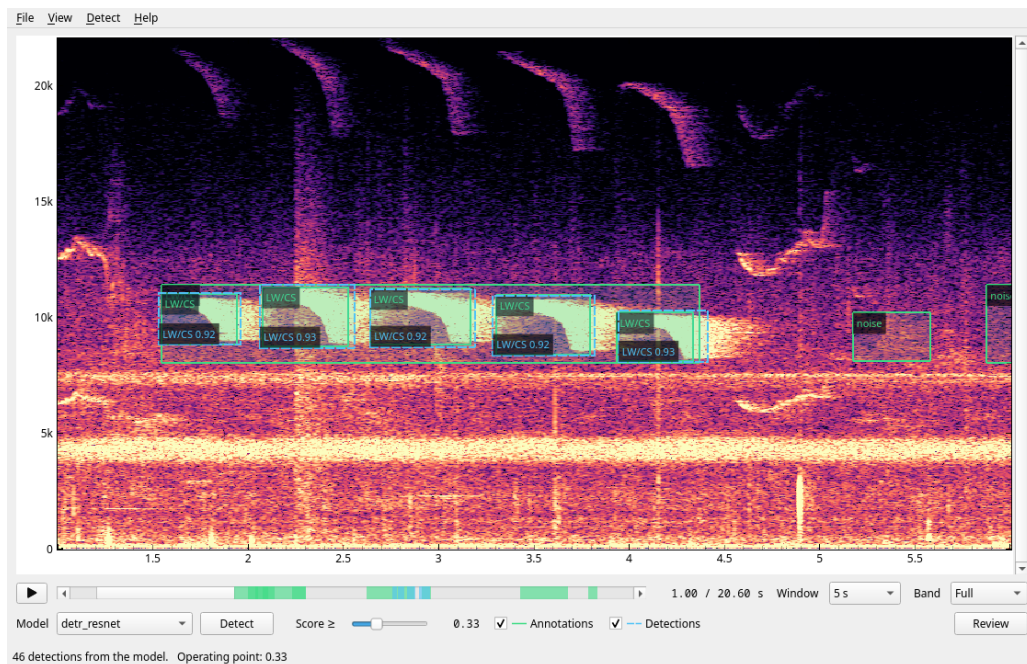


Primate Vocalization Detector

User Guide



Version 0.1.5 · Windows 10/11, 64-bit

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1 Introduction

The program finds monkey calls in audio recordings. It draws a *box* around each call on the spectrogram and tells you the species, the call type and a *score* (how sure it is).

The boxes are saved as a **Raven selection table**: a text file that opens in Raven Pro, Excel or the program itself. For a recording `name.wav` you get `name.detections.txt` next to it.

You get two programs:

- `viewer.exe` — the window. Open a recording, run the model, listen, review the boxes one by one, process a whole folder.
- `detect.exe` — the command line. Process folders without a window, or from a script.

2 Installation

1. Download from <https://github.com/nhrot-fc/thesis-primate/releases>:
 - `detector-0.1.5-win64-cpu.zip` (any PC) **or** `detector-0.1.5-win64-cuda.zip` (PC with an NVIDIA graphics card);
 - at least one model: `detector-0.1.5-model-name.zip`.
2. Unzip the `win64` zip into a short path, for example `C:\detector\`.
3. Unzip the model zip into **the same folder**, next to `viewer.exe`.

Nothing else to install: Python and all the libraries are inside. No internet needed.

Windows	10 or 11, 64-bit
Memory	8 GB (16 GB with a graphics card)
Disk	3 GB free (<code>cpu</code>) or 7 GB (<code>cuda</code>)
Graphics card	optional; NVIDIA with 4 GB and driver 570 or newer

3 Execution

Double-click `viewer.exe`. To open a recording right away, drag it onto `viewer.exe`.

The first time:

- Windows says *Windows protected your PC*. Click **More info**, then **Run anyway**. It happens only once.
- The launch is slow: the antivirus checks the new files.

Every time:

- The status bar says *Loading detection engine* for a few seconds. You can open a recording meanwhile. **Detect** turns on when it says *Ready*.
- If the program does not open, read `viewer.log`, next to `viewer.exe`.

For the command line, see section 6.

4 Adding a model

A model is a folder inside `models\`. There are three models: Faster R-CNN (most accurate, slowest), YOLO26s (fastest) and AST-Deformable DETR (in between).

1. Download its zip, `detector-0.1.5-model-name.zip`.

2. Unzip it into the program folder (next to `viewer.exe`).
3. Pick it in the **Model** list, under the spectrogram (the **Batch** view has the same list).

With one model installed, it is already picked.

5 Viewer

5.1 The window

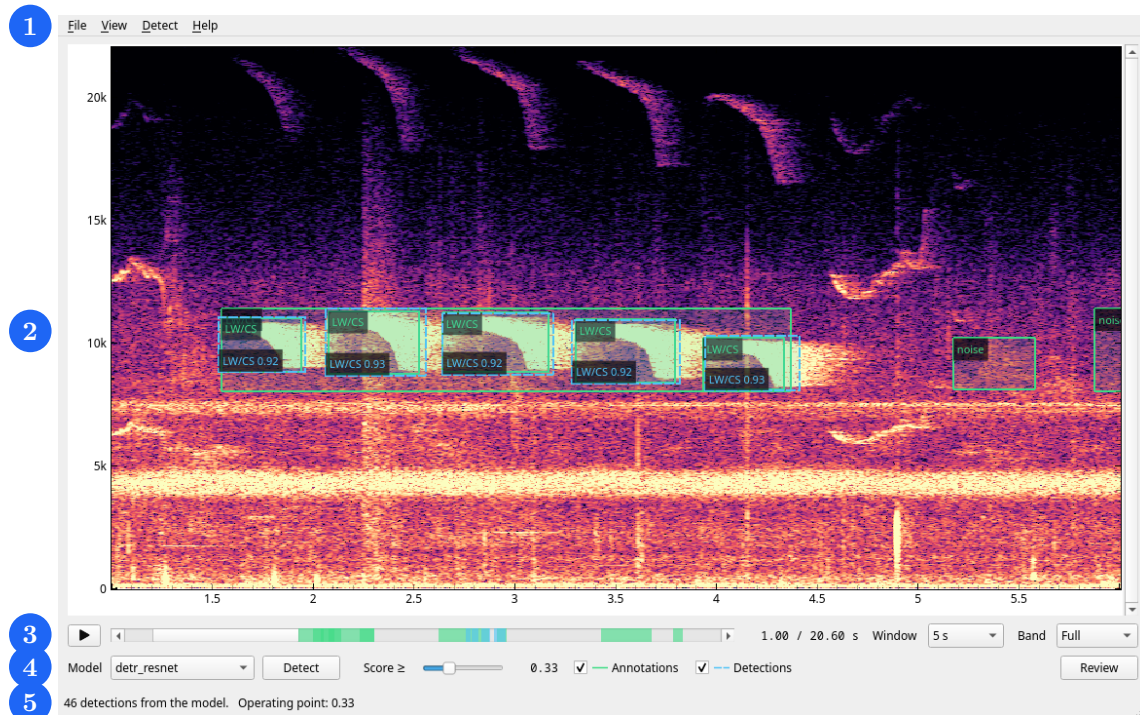


Figure 1: A recording with annotations (green) and detections (blue, with score).

- 1 **Menus.** Everything the program does, with its keyboard shortcut: open, save, the two views (**Spectrogram**, **Batch**), zoom, **Boxes**, **Settings**.
- 2 **Spectrogram.** Time left to right, frequency bottom to top.
- 3 **Transport.** Play, the time bar (the marks are the boxes), **Window** (how many seconds you see) and **Band** (how many Hz you see).
- 4 **Detection.** The **Model**, **Detect**, the **Score** filter, the two layers and **Review**.
- 5 **Status bar.** What just happened, and what is running.

5.2 Open a recording

Drag a WAV, FLAC or MP3 file onto the window, or **File** > **Open audio...** (**Ctrl** + **O**).

If the recording already has a `.detections.txt` next to it, it opens too.

To see a Raven table made by hand: **File** > **Open annotations...** (**Ctrl** + **T**). It shows in green.

5.3 Detect

1. Press **Detect**, under the spectrogram (**Ctrl** + **R**).
2. Wait. The status bar shows the progress.

3. The boxes appear in blue with their score.



Figure 2: Below the spectrogram: **Model** and **Detect**, the **Score** filter, the layers and **Review**.

Score $\geq$ hides the boxes with a lower score. It starts at the best value for the model. Move it left to see more boxes, right to see fewer.

View > **Boxes** (Ctrl + E) opens the list of boxes. Click a row to go to that box; **N** and **P** go to the next and previous one. Press **Del** to remove it.

5.4 Review

Review shows the detections one at a time so you can check them.

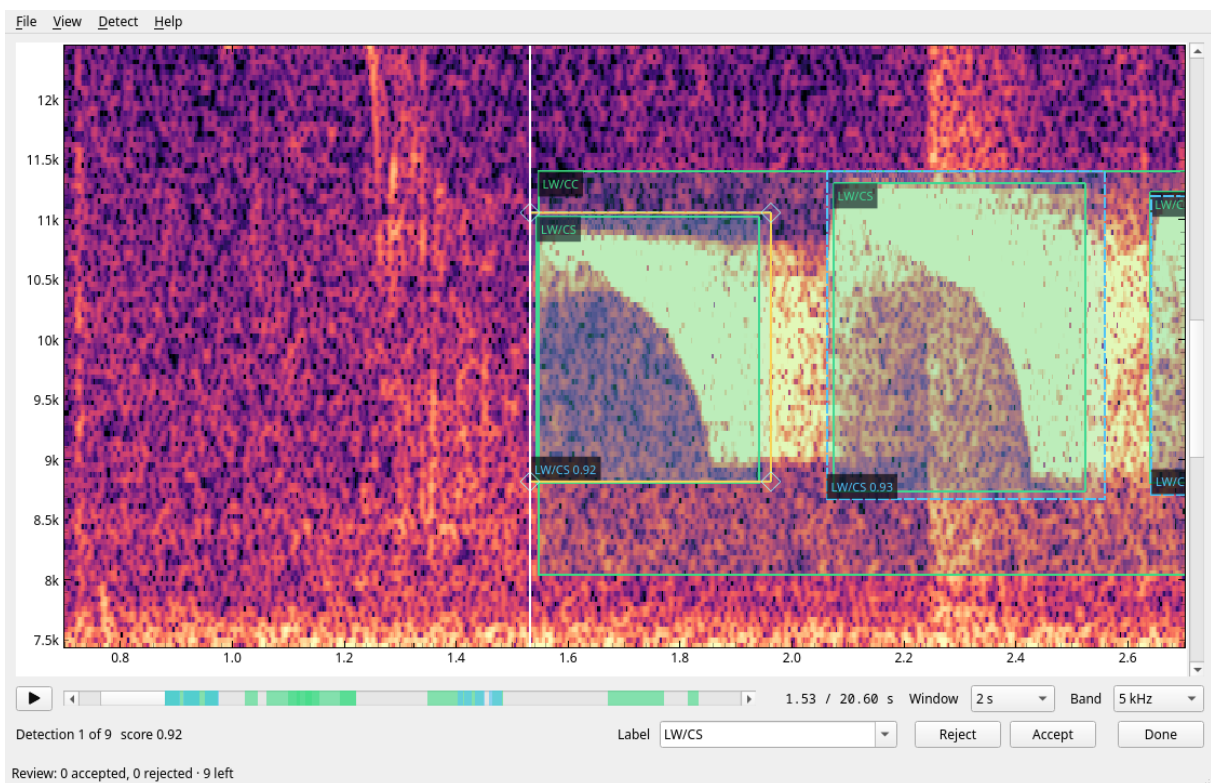


Figure 3: Review: the box is zoomed in and has handles.

For each box:

- Press **Space** to hear it.
- Drag the box or its corners if it is not well placed.
- Change the **Label** if the species or call is wrong.
- Press **A** to **Accept** or **R** to **Reject**. The next box appears.
- **N** and **P** skip forward and back. **Esc** leaves.

Accepted boxes go to the green layer. Your progress is saved: if you close the program, the review continues where you left it.

5.5 Save

- **File** » **Save annotations table...**: the green boxes (accepted ones).
- **File** » **Save detections table...**: the blue boxes you can see.
- **File** » **Save image...** (**Ctrl** + **S**): the screen as a PNG.

Tables are Raven files. In Raven Pro, open the audio and then **File** » **Open Selection Table**.

5.6 Batch: a whole folder

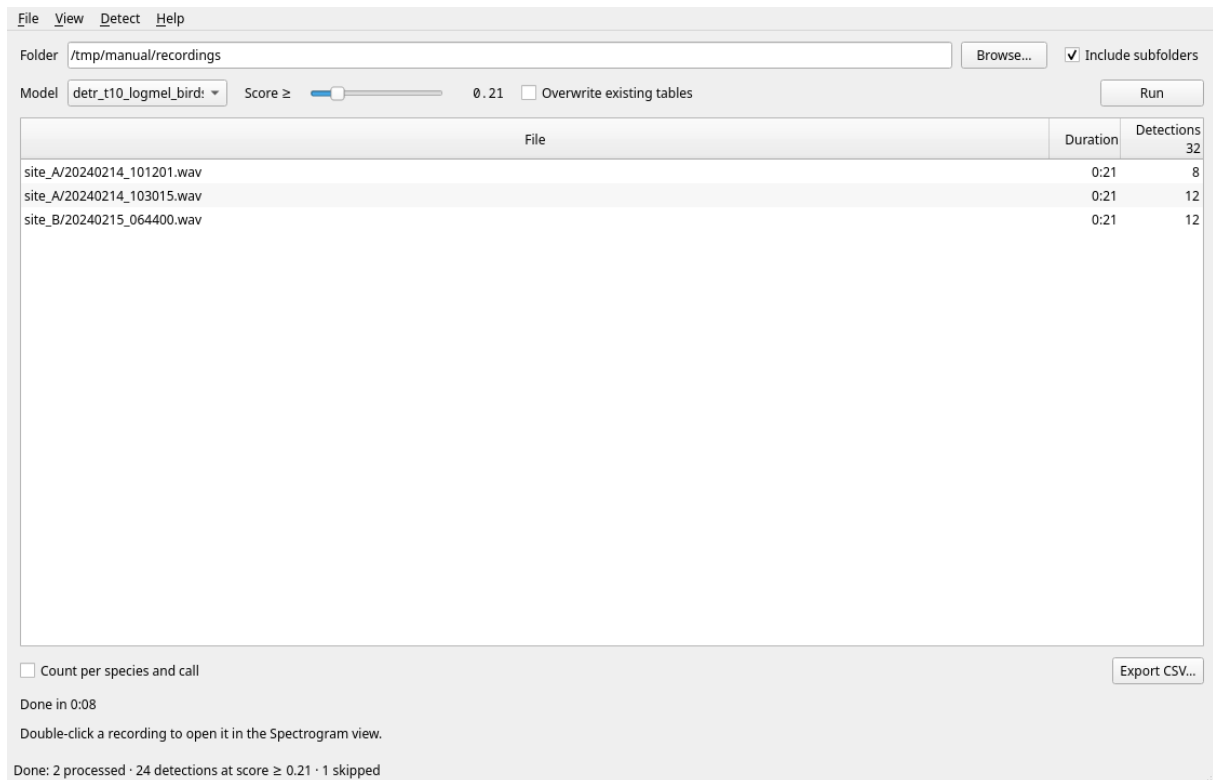


Figure 4: Batch after a run: each recording with its detections at the chosen score.

1. **View** » **Batch**.
2. Drag a folder onto the window, or **Choose a folder...** (**File** » **Open folder...**, **Ctrl** + **↑** + **O**, does both).
3. Click **Run**.

Every recording gets its `.detections.txt`. Files that already have one are skipped (tick **Overwrite existing tables** to redo them). **Stop** stops the run; finished files are kept. Double-click a file to open it in the **Spectrogram** view.

The tables keep every detection, and **Detections** counts the ones at or above **Score**; moving it recounts without running the model again (the folder total is under the column name). Tick **Count per species and call** for one column per class of the model. **Export CSV...** saves one row per recording, with every class.

5.7 Move and zoom

Mouse wheel	move in time
Ctrl + wheel, or Ctrl + 1 / Ctrl + 3	zoom in time
⇧ + wheel, or Ctrl + ⇧ / Ctrl + ⇩	zoom in frequency
⇧ ⇩	move the frequency band; F shows all
← → , PgUp PgDn	step, page
Home End	start, end
N P	next, previous box (its row is selected in Boxes)
Space	play / pause
Click on the spectrogram	move the playhead

F1 shows all shortcuts inside the program.

6 Detector CLI

detect.exe does the same as **Batch**, from the command line. Open **cmd** or PowerShell in the program folder and run:

```
detect.exe --model models\yolo26s_v2 D:\recordings
```

-model	Which model. Without it, the only installed model is used.
-score S	Minimum score to keep a box. Default: the best value for the model.
-overwrite	Redo files that already have a table.
-device cpu	Do not use the graphics card.

You can give several folders or files. What you see:

```
Loading detection engine...
[INFO] yolo26s_v2 on cuda | score >= 0.10 | 3 recordings
[INFO] [1/3] 20240214_101201.wav already has a table; --overwrite redoes it
[INFO] [2/3] 20240214_103015.wav: 14 detections in 1 s -> 20240214_103015.detections.txt
[INFO] [3/3] 20240215_064400.wav: 14 detections in 1 s -> 20240215_064400.detections.txt
[INFO] Done. Open each audio in Raven with its .detections.txt table
```

A script that runs two folders overnight, **run.bat**:

```
cd /d C:\detector
detect.exe --model models\frcnn D:\field\2024-02 D:\field\2024-03 > run.log
```

7 Audio settings

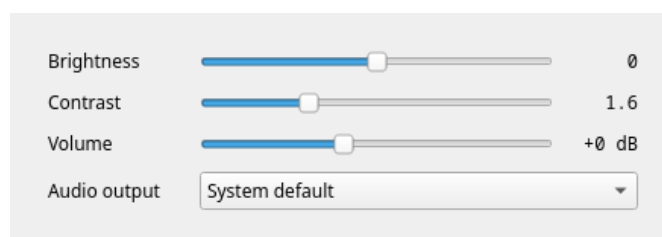


Figure 5: **View** **Settings...**

- **Audio output**: where the sound plays (speakers, headphones).
- **Volume**: raise it for distant calls. Field recordings are quiet.

- **Brightness** and **Contrast**: how the spectrogram looks.

8 If something goes wrong

- **No sound**:  → **Audio output**, pick your device.
- **Detect is grey**: wait for *Ready* in the status bar.
- **No model in the list**: the model zip must be unzipped next to `viewer.exe`, not inside `models\`.
- **Very slow**: use YOLO26s, or the `cuda` package on a PC with an NVIDIA card.
- **The program closes**: `viewer.log`, next to `viewer.exe`, says why.